

THEORY

1. What is Logistic Regression and why is it suitable for classification?

Answer:

Logistic Regression is a supervised machine learning algorithm used for solving classification problems where the target variable belongs to different categories or classes. Although its name contains "regression," it is mainly used for classification tasks such as predicting whether a customer will default on a loan, whether an email is spam or not, or whether a transaction is fraudulent. It uses a mathematical function called the sigmoid (logistic) function that converts input features into a probability value between 0 and 1. Based on a predefined threshold, usually 0.5, the probability is converted into a class label.

Logistic Regression is suitable for classification because it is simple, efficient, easy to interpret, and provides probability estimates rather than only final predictions. It helps understand the relationship between independent variables and the probability of an event occurring. It is widely used in risk prediction, healthcare diagnosis, customer behavior analysis, and financial decision-making because it performs well when the relationship between input features and output classes is approximately linear.

Logistic Regression is a **supervised machine learning algorithm** used for solving **classification problems** where the output variable belongs to one or more predefined categories. It is one of the most widely used statistical learning algorithms for predicting the probability of an event occurring based on input features.

Although the name contains the word "**Regression**," Logistic Regression is not used for predicting continuous numerical values like Linear Regression. Instead, it is mainly used for **classification tasks** such as:

- Predicting whether a customer will default on a loan or not.
- Detecting whether an email is spam or not.
- Predicting whether a transaction is fraudulent or genuine.
- Medical diagnosis such as whether a patient has a disease or not.

Logistic Regression works by finding the relationship between **independent variables (input features)** and a **dependent categorical variable (target class)**. It calculates a weighted combination of input features and applies a special mathematical function called the **Sigmoid Function**.

The sigmoid function converts any input value into a probability value between **0 and 1**.

2. Explain classification performance metrics and why accuracy alone is insufficient.

Classification performance metrics are evaluation techniques used to measure how effectively a machine learning classification model predicts different classes. These metrics compare the actual output values with the predicted values generated by the model.

The main classification evaluation metrics include:

- **Accuracy**
- **Precision**
- **Recall**
- **F1-score**
- **True Positive Rate (TPR)**
- **False Positive Rate (FPR)**
- **AUC-ROC**

Answer:

Classification performance metrics are evaluation measures used to determine how effectively a machine learning classification model predicts different classes. These metrics help analyze the strengths and weaknesses of a model by comparing predicted values with actual values. Important classification metrics include accuracy, precision, recall, F1-score, True Positive Rate (TPR), False Positive Rate (FPR), and AUC-ROC.

Accuracy represents the percentage of total correct predictions made by the model. However, accuracy alone is insufficient because it does not provide information about the type of errors made by the model. In datasets with class imbalance, accuracy can be misleading. For example, in credit risk prediction, if 95% of customers are low risk and only 5% are high risk, a model that predicts every customer as low risk can achieve 95% accuracy but completely fail to identify risky customers. Therefore, additional metrics such as recall, precision, and F1-score are necessary to evaluate the model's real performance, especially when detecting important minority classes.

3. Define Type-I Error and Type-II Error in the context of risk prediction.

Answer:

In classification problems, errors occur when the model's prediction does not match the actual outcome. These errors are categorized as Type-I Error and Type-II Error. Type-I Error, also called a False Positive (FP), occurs when the model predicts a positive result when the actual result is negative. In risk prediction, this means the model incorrectly identifies a low-risk customer as a high-

risk customer. This may result in unnecessary rejection of loans, loss of potential customers, or additional verification costs.

Type-II Error, also called a False Negative (FN), occurs when the model predicts a negative result when the actual result is positive. In risk prediction, this means the model identifies a high-risk customer as low-risk and approves their application. This error is usually more dangerous in financial applications because it can lead to loan defaults, financial losses, and increased business risk. The balance between Type-I and Type-II errors depends on the business objective and the cost associated with each mistake.

In classification problems, errors occur when the predicted output of a machine learning model does not match the actual outcome. These errors are mainly classified into two types:

1. Type-I Error (False Positive)
2. Type-II Error (False Negative)

Type-I Error (False Positive):

Type-I Error occurs when the model predicts a positive result, but the actual result is negative.

In simple words:

The model identifies something as risky, but it is actually safe.

Example in credit risk:

A customer is actually low-risk, but the model predicts that the customer is high-risk.

Consequences:

- Good customers may be rejected.
- Business opportunities may be lost.
- Customer satisfaction may decrease.
- Extra verification costs may increase.

Type-II Error (False Negative):

Type-II Error occurs when the model predicts a negative result, but the actual result is positive.

In simple words:

The model identifies something as safe, but it is actually risky.

Example in credit risk:

A high-risk customer is classified as low-risk.

Consequences:

- Loan may be approved for risky customers.
- Higher chances of default.
- Financial losses may occur.
- Business risk increases.

In financial applications, Type-II Error is usually considered more dangerous because it allows risky customers to pass through the system.

4. Explain Precision, Recall, F1-Score, TPR, and FPR.

Answer:

Precision measures the correctness of positive predictions made by a classification model. It represents the percentage of predicted positive cases that are actually positive. A high precision value means the model produces fewer false positive predictions. It is important in situations where false alarms are costly.

Recall, also known as Sensitivity or True Positive Rate (TPR), measures the ability of a model to correctly identify all actual positive cases. It shows how many positive instances were successfully detected by the model. In risk prediction, high recall is important because missing high-risk cases can create serious consequences.

F1-Score is a combined evaluation metric that balances precision and recall. It is calculated as the harmonic mean of precision and recall. F1-score is especially useful when dealing with imbalanced datasets because it considers both false positives and false negatives.

True Positive Rate (TPR) represents the proportion of actual positive cases that are correctly classified as positive. It indicates the model's ability to detect positive cases.

False Positive Rate (FPR) represents the proportion of actual negative cases incorrectly classified as positive. A lower FPR indicates fewer incorrect positive predictions and better model reliability.

5. What is AUC-ROC and how does it help in evaluating classifiers?

Answer:

AUC-ROC stands for Area Under the Receiver Operating Characteristic Curve. It is a performance evaluation metric used to measure how well a classification model can distinguish between different

classes. The ROC curve is created by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at different classification thresholds.

The AUC value represents the overall ability of the model to separate positive and negative classes. The value ranges from 0 to 1. A model with an AUC close to 1.0 indicates excellent classification ability, while an AUC value around 0.5 represents a model performing no better than random guessing. AUC-ROC is useful because it evaluates model performance across multiple thresholds instead of relying on a single prediction threshold. It is commonly used in banking, healthcare, fraud detection, and risk assessment applications where choosing the correct classification threshold is important.

6. Why does imbalanced data create problems in classification models?

Answer:

Imbalanced data occurs when the distribution of classes in a dataset is unequal, where one class has significantly more samples than another class. For example, in credit risk prediction, the number of low-risk customers may be much higher than high-risk customers. This imbalance can cause machine learning models to become biased toward the majority class because they learn to predict the most common outcome.

As a result, the model may achieve high accuracy while performing poorly on the minority class, which is often the most important class to identify. For example, a fraud detection model may correctly classify most normal transactions but fail to detect fraudulent transactions. To overcome this issue, techniques such as oversampling (SMOTE), under sampling, class weighting, and threshold adjustment can be applied. Additionally, evaluation metrics such as recall, F1-score, precision, and AUC-ROC should be considered instead of accuracy alone to obtain a more reliable understanding of model performance.